MILITARY SPECIFICATION

BITS, SCREWDRIVER, RIBBED
GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of ribbed screwdriver bits and bit assemblies used for installation and removal of recess drive fasteners.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-40	Packaging and Packing of Hand Tools
QQ-T-570	Tool Steel, Alloy

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 93), Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5120

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables of Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1312	Fastener Test Methods

(See Supplement 1 for applicable Specification Sheets.)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual bit shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.5).

3.3 Material. The material shall be as specified on the applicable specification sheet.

3.4 Dimensions. The dimensions and contour shall be in accordance with the applicable specification sheet. The dimensions and contour gaging test shall be conducted per 4.5.1.

3.5 Surface characteristics.

3.5.1 Finish. All surfaces shall be free of cracks (see 6.4) regardless of depth. The driving blade and drive adapter surfaces shall be free of discontinuities, such as laps (see 6.4) greater in depth than 2 percent of the thinnest material at the discontinuity location, and inclusions (see 4.5.2.1).

3.5.2 Smoothness. The driving blade surface shall be not more than 125 microinches. All other surfaces shall be not more than 250 microinches (see 4.5.2.2).

3.5.3 Surface treatment. The bit shall be free of material permanently deposited on the surface (such as a plating or coating (see 6.4)). A rust preventive coating shall be applied (see 4.5.2.3).

3.6 Hardness. The hardness range of the bit shall be as specified on the applicable specification sheet (see 4.5.3).

3.7 Performance.

3.7.1 Torque fatigue. Using the torque values in the applicable specification sheet, the bits shall be subjected to not less than 50 cycles in both the clockwise and counterclockwise direction (see 4.5.4).

3.7.2 Minimum ultimate torque. The bits shall be subjected to the application of the minimum ultimate torque values in the applicable specification sheet without permanent deformation or fracture (see 4.5.5).

3.8 Identification of product. The bit shall be marked in accordance with the applicable specification sheet.

3.9 Workmanship. The bit shall be free from burrs, scale, excessive seams and other defects which would interfere with their intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspections. First article inspections shall be as specified in Table I.

TABLE I. First article inspections.

Test	Requirement Paragraph	Test Paragraph	Quantity of Each Size	Level	AQL
Examination of product	3.3, 3.4	4.5.1, 4.5.1.1	25	II	1.0%
Surface characteristics	3.5	4.5.2	25	II	1.0%
Hardness	3.6	4.5.3	25	II	1.0%
Torque Fatigue	3.7.1	4.5.4	25	S3	1.5%
Minimum Ultimate Torque	3.7.2	4.5.5	10	S3	1.5%

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4.3.1 First article samples. Unless otherwise specified, as soon as practicable after award of a contract or order, the manufacturer shall submit samples as specified in the contract or order. The samples shall be representative of the construction, workmanship, components and materials used during production.

4.3.2 Certified test report. The manufacturer shall furnish a certified test report showing that the manufacturer's product satisfactorily conforms to this specification. The test report shall include actual results of the tests specified herein. When this report is submitted, it shall be accompanied by a dated drawing that completely describes the manufacturer's product. The drawing shall specify all dimensions and tolerances, composition of materials selected, forming process (machined, stamped, forged or drawn), and the Rockwell hardness and heat treatment. The drawing shall also specify the manufacturer's part number for each size.

4.4 Quality conformance inspections. Quality conformance inspections shall consist of the tests specified in Table II.

TABLE II. Quality conformance inspections.

Test	Requirement Paragraph	Test Paragraph	Level	AQL
Examination of product	3.3, 3.4	4.5.1	II	1.0%
Surface characteristics	3.5	4.5.2	II	1.0%
Hardness	3.6	4.5.3	SI	2.5%
Torque Fatigue	3.7.1	4.5.4	S3	1.5%
Minimum Ultimate Torque	3.7.2	4.5.5	S3	1.5%

4.4.1 Sampling.

4.4.1.1 Random samples. Random samples for inspection and testing shall be selected from each inspection lot in accordance with MIL-STD-105. Inspection levels and AQLs shall be as specified in Table II. Failure of any single sample to meet the requirements of this specification shall be cause for rejection.

4.4.1.2 Production lot. A production lot shall consist of bits which are the same type, class and size, fabricated from a single mill heat, heat treated and produced as one continuous run.

4.4.1.3 Inspection lot. An inspection lot shall consist of bits from a single production lot.

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4.5 Methods of inspection.

4.5.1 Dimensions. Sample bits shall be inspected for dimensional accuracy using conventional optical measuring equipment (a micrometer or calipers) and a gage as specified in 4.5.1.1 Noncompliance with the dimensions specified on the applicable specification sheet or drawing shall be cause for rejection.

4.5.1.1 Gage. The sample bit shall meet the "Go - Not Go" criteria established by gages for driver configuration specified in the applicable specification sheet. Noncompliance with the gage shall constitute rejection of the lot.

4.5.2 Surface characteristics.

4.5.2.1 Finish. The sample bit shall be microscopically examined at a magnification of not less than 10X. Presence of any cracks, inclusions or discontinuities greater than the limits specified in 3.5.1 shall be cause for rejection.

4.5.2.2 Smoothness. Sample bits shall be inspected in accordance with ANSI B46.1 for surface smoothness.

4.5.2.3 Surface treatment. Sample bits shall be visually inspected for conformance to the requirements specified in 3.5.3.

4.5.3 Hardness. Sample bits shall be tested in accordance with ASTM E 18.

4.5.4 Torque fatigue test.

4.5.4.1 Test sample. The bits tested for torque fatigue shall have previously met the dimensional, surface characteristics, and hardness requirements of 3.4, 3.5, 3.6. The recesses shall be free of foreign matter.

4.5.4.2 Test procedure. Testing shall be in accordance with MIL-STD-1312, Test 25, except as follows:

a. Use the appropriate test recess specified in the applicable specification sheet.

b. The end load shall be not less than 20 inch-pounds and not greater than 20.5 inch-pounds.

c. Use the torque load values in the applicable specification sheet.

d. The bit shall be subjected to not less than 50 torque cycles in both the clockwise and counterclockwise direction.

4.5.4.3 Failure. Upon completion of 50 cycle test, failure to meet the dimensional and finish requirements of 3.4 and 3.5 is cause for rejection of the production lot.

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4.5.5 Minimum ultimate torque test.

4.5.5.1 Test sample. The bits shall have previously met the dimensional surface characteristics and hardness requirements of 3.4, 3.5, and 3.6. The recesses shall be free of foreign matter.

4.5.4.2 Test procedure. Testing shall be in accordance with MIL-STD-1312, Test 25, except as follows:

a. Use the appropriate test recesses specified in the applicable specification sheet.

b. The end load shall be not less than 20 inch-pounds and not greater than 20.5 inch-pounds.

c. The driving blade shall be aligned with the recess axis within $\pm 3°$.

d. Using a calibrated torque wrench, apply the minimum ultimate torque load valves in the aplicable specification.

e. Apply the specified torque once in the clockwise direction and once in the counterclockwise direction.

f. If failure or cam-out of the test recess occurs three times, check the hardness requirements of the test recess.

4.5.5.3 Failure. Fracture or permanent set of the bit shall be cause for rejection of the production lot. Failure or cam-out of the test recess shall not constitute a bit failure.

4.5.5.4 Retesting. Bits failing to pass the minimum torque tests shall not be used for further tests. Retesting shall be conducted using new bits and recesses.

4.5.6 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of 5.1.

5. PACKAGING

5.1 Packaging requirements. Bits covered by this specification shall be packaged in accordance with PPP-P-40. Levels of packaging shall be as specified in the contract or order (see 3.8 and 6.2).

5.2 Marking. In addition to any special marking specified in the contract or purchase order, marking shall be in accordance with the applicable specification sheet and MIL-STD-129.

6. NOTES

6.1 Intended use. Bits shall be capable of being used with selected hand and power wrenches in the installation and removal of recess drive fasteners. Bits must have rust preventive coating removed prior to use.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Bit identification (see applicable specification sheet).

c. Bit shank size and length (see applicable specification sheet).

d. Additional tests required.

e. Levels of preservation packaging and packing (see 5.1).

f. Special marking (see 5.2).

6.3 Inspection and acceptance. Acquisition documents shall also include one of the following requests:

a. Direct government procurement: Inspection and acceptance shall be at the source of manufacture or supply for direct government procurements.

b. Contractor procured bits intended for delivery to the Government: Inspection and acceptance shall be at the source or at the contractor's receiving point.

6.4 Definitions.

a. Bits. A bit is a tool with an appropriately shaped driving blade (see 6.4b) used to apply torque to the recess drive element of a screw or bolt.

b. Driving blade. The driving blade is that portion of the bit that engages the recess drive element in the screw or bolt.

c. Drive adapter. The drive adapter is the portion of the bit that engages the wrench.

d. Laps. Laps are material discontinuities (folded metal) that do not cross grain boundaries.

e. Cracks. A crack is a clean, crystalline break passing through grain or grain boundary.

f. Plating and coating. Plating is electro-chemical deposition of material (such as cadmium, chromium, zinc, etc.); coating is a mechanical deposition of material (such as paint, organics, suspension containing metals, etc.). Coating and plating do not include chemical processes such as anodize, black oxide or black phosphate (see 3.5.3).

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6.5 First article. When a first article inspection is required, the items should be a standard production item from the contractor's current inventory as specified in 4.3.1. The first article shall consist of 25 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Preparing Activity:
Navy - AS
(Project No. 5120-NC11)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85643	BITS, SCREWDRIVER, RIBBED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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